Feature and Implementation Reference

Architecture, API, data model and analytical capabilities

Portfolio context

Developed during my IT internship at Murphy Ben International (MBI) as an internship/internal experimental project for collecting and analyzing broadcast signal reports. I was responsible for the design and implementation of the application, working under the guidance of a senior member of the IT team.

This repository is a portfolio representation of the project. Sensitive configuration, internal information, production data, credentials, and company-specific deployment details have been excluded.

1. Architecture

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Browser interfaces
-> Microsoft IIS (HTTPS 443)
-> static HTML/CSS/JavaScript
-> /api/* through URL Rewrite and ARR
-> Node.js/Express on 127.0.0.1:3000
-> JSON runtime stores
```

2. Core features

Category	Capabilities
Capture	GPS, accuracy, station/channel context, comments and configurable choices
Identity	Server-generated PUB and INC IDs with separate continuous sequences
Reliability	Submission-key duplicate checks and offline Field queue
Analysis	Haversine distance, observation scores, nominal EIRP and RF reference
Operations	Status, assignment, filters, audit and health checks
Reporting	CSV, print, maps and Google Earth KML

3. API surface

Method	Route	Purpose
GET	/api/health	Runtime health
GET/PUT	/api/config	Read or update configuration
POST	/api/rf-estimate	Generate an RF reference
GET/POST	/api/incidents	Read or create incidents
PATCH/DELETE	/api/incidents/:id	Update or delete an incident
GET	/api/public-incidents	Public incident views
GET	/api/analysis/incidents	Combined incident views
GET	/api/analysis/summary	Combined analytical summary
GET	/api/analysis/incidents/export.csv	Combined CSV export

4. Data model

Runtime file	Purpose
incidents.json	Field incidents
public_incidents.json	Public incidents
config.json	Application configuration
audit.json	Administrative and API activity
id_sequences.json	Persistent PUB and INC counters

Incident identity

```
PREFIX + REVERSED FULL DATE + HHMMSS + CONTINUOUS SEQUENCE
```

5. RF and location processing

- Haversine distance between report and configured station.
- RF transmitter-power normalization and nominal EIRP.
- Reference field-strength calculation in dB microvolts per metre.
- Quality, stability and service-condition scoring.
- Terrestrial horizon-limited reception-radius reference.

> The RF model does not include terrain, buildings, vegetation, interference, diffraction, directional patterns, feeder loss, fading or measured propagation calibration.

6. Configuration ownership

- Design controls appearance.
- Menu Management controls navigation.
- Component Management controls sections and choice definitions.
- Form Builder controls field behavior and custom fields.
- Dashboard Layout controls widgets.
- Stations, channels and GPS settings are shared technical authorities.